

AI Career Navigator

System Architecture, Entity-Relationship Schema & Career Readiness Engine Documentation

Target Platform: AI Career Navigator	Database: PostgreSQL 16 (11 Tables)
Runtime: Python 3.14 / React 18 + Vite	Regression Status: 471/471 Tests Passing (100%)
Git Baseline: commit e88b7e9 (main)	Build Status: Frontend Vite Production Build: Exit Code 0

1. Executive Summary & Core Mission

AI Career Navigator is an evidence-based, deterministic career intelligence platform designed for undergraduate engineering and computer science students. Rather than relying on black-box heuristics or generative chatbots, the system synthesizes academic degree benchmarks, live industry vacancy requirements, practical technical tasks, and explainable ROI metrics to answer four primary student questions: *What should I do TODAY?*, *What should I do NEXT?*, *What is blocking me?*, and *Am I becoming job-ready?*

2. Final System Architecture

The system employs a strict 4-tier modular architecture cleanly segregating client-side user experience, API controllers, deterministic analytical service engines, and relational persistence:

Layer	Components & Technologies	Responsibilities
Frontend Client	React 18.3, Vite 5.4, Tailwind CSS 3.4, Lucide React Icons	Responsive dashboards, radial readiness gauges, blocker drawers, 5-day schedules, ATS parser UI.
API Gateway & Routes	Flask 3.1, Flask-JWT-Extended, REST Blueprints (/api/*)	Request validation, JWT token issuance, career endpoints, simulation payload ingestion.
Intelligence Services	12 Specialized Service Modules (SkillGap, ROI, Adaptive, Readiness, etc.)	Deterministic in-memory calculations, dynamic weight redistribution, 7-dimension blocker diagnosis.
Data & Persistence	PostgreSQL 16, SQLAlchemy 2.0 ORM, psycopg2-binary	11 persistent relational tables preserving users, profiles, canonical skills, and resources.

Deterministic Information Flow:

User Profile & Target Career → SkillGapService (1-5 Level Gaps) → AdaptiveLearningService (Curriculum Sequencing) → Multi-Evidence Triangulation (Skill Coverage 40%, Practical 25%, Portfolio 20%, Interview 15%) → Dynamic Weight Redistribution → CareerReadinessService → JSON API → React Executive Dashboard.

3. Final Entity-Relationship (ER) Schema & Database Specification

Database: career_navigator (PostgreSQL 16 on localhost:5432). The schema contains **exactly 11 tables** with zero DDL mutations across all phases:

#	Table Name	Primary Key	Foreign Keys	Key Columns & Purpose
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1	users	id	None	name, email (unique), password_hash. User identity and authentication.
2	student_profiles	id	user_id → users.id	education, specialization, graduation_year, cgpa. Academic background.
3	career_preferences	id	user_id → users.id	target_role, preferred_domain, experience_level. Student aspirations.
4	data_sources	id	None	source_name, source_type, website, license, source_version. Audit provenance.
5	canonical_skills	id	None	canonical_name, normalized_name (unique), skill_type, description. Canonical registry.
6	skill_aliases	id	canonical_skill_id, source_id	alias_name, normalized_alias. Normalization mapping for syntax variations.
7	skills	id	user_id, canonical_skill_id	skill_name, proficiency (1-10 raw). Verified student skill inventory.
8	careers	id	None	title, domain, description. The 10 canonical IT career tracks.
9	career_skills	id	career_id, canonical_skill_id	skill_name, required_level (1-5), importance (1-5). Track requirements.
10	learning_resources	id	canonical_skill_id	title, resource_type, provider, url, difficulty, certification_available.
11	user_learning_progresses	id	user_id, resource_id, canonical_skill_id	status, progress_percentage, notes, started_at, completed_at. Completion audit.

4. Phase 12.5 Career Readiness & Personalized Action Plan Engine

Phase 12.5 operationalizes career navigation into four deterministic modules:

Student Question	Engine Output Field	Algorithmic Mechanism
What should I do TODAY?	today	Identifies single highest-leverage atomic action (30-90 min). Prioritizes unblocking prerequisites, followed by highest ROI competency study, practical tasks, or capstone work with explicit 'Why Now' reasoning.
What should I do NEXT?	next_actions & weekly_plan	Generates 4-pillar progression (LEARN → PRACTICE → BUILD → PREPARE) and customizable 5-day schedule calibrated to 5h, 10h, 15h, or 20h weekly budgets with exact sum preservation.
What is blocking me?	blockers	7-dimension root-cause diagnosis: Prerequisite, Critical Gap, Practical Evidence, Portfolio Capstone, Interview Articulation, Academic Blind-Spot, and Live Market Vacancy requirements.
Am I becoming job-ready?	readiness	Multi-evidence composite score (0-100) dynamically normalized across established evidence layers. Maps student into 3 educational bands and 5 lifecycle milestone states.

Multi-Evidence Dynamic Normalization Algorithm:

Base Weights: Skill Coverage (40%), Practical (25%), Portfolio (20%), Interview (15%).

When optional diagnostic layers are unestablished, weights dynamically normalize over established layers ■:

$$W_{total} = \sum_{k \in \blacksquare} W_k^{base} \quad | \quad w_k = W_k^{base} / W_{total} \quad | \quad \text{Overall Score} = \sum_{k \in \blacksquare} (w_k \times S_k)$$

Guarantee: Advanced students are never penalized with 0% for unattempted mock tests.

Metric / Scale	Thresholds & States	Pedagogical Interpretation
Educational Bands	FOUNDATION_REQUIRED (<50) DEVELOPING (50-74.9) PLACEMENT_READY (≥75)	Formative academic grading representing syllabus alignment, hands-on diagnostics, and hiring benchmark satisfaction.

5-Tier Milestones	1. NOT_READY (<40 or prereq blocked) 2. FOUNDATION_BUILDING (40-59.9) 3. PROJECT_APPLICATION (60-74.9) 4. INTERVIEW_ACTIVE (75-84.9) 5. PLACEMENT_READY (≥85)	Lifecycle state progression identifying current focus: unblocking prerequisites → diagnostic tasks → capstone build → mock interview → active vacancy applications.
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5. Final Technology Stack

Layer	Technology	Version / Specification
Language & Runtime	Python	3.14 (virtualenv: backend/venv)
Backend Framework	Flask	3.1.3 (Werkzeug 3.1.8, Jinja2 3.1.6)
Database ORM	SQLAlchemy / Flask-SQLAlchemy	2.0.52 / 3.1.1 (psycopg2-binary 2.9.12)
Security & Auth	Flask-JWT-Extended	4.7.4 (PyJWT 2.13.0, PBKDF2 SHA-256)
Database Server	PostgreSQL	16.x (localhost:5432, db: career_navigator)
Frontend Runtime	React	18.3.1 (react-dom 18.3.1)
Frontend Tooling	Vite	5.4.21 (@vitejs/plugin-react 4.3.1)
Routing	react-router-dom	6.26.2
CSS Architecture	Tailwind CSS	3.4.10 (PostCSS 8.4.45, Autoprefixer 10.4.20)
Icon Library	lucide-react	0.441.0
Report Engine	ReportLab	5.0.1 (PDF Generation)

6. Implemented Features vs. Future Scope Roadmap

Category	Feature Module	Implementation Status & Details
Authentication & Profile	JWT Auth & Student Profiles	Implemented & Verified: Secure registration, JWT token lifecycle, student academic profile CRUD, career preferences.
Competency Gaps	Phase 10 Skill Gap Engine	Implemented & Verified: 1-5 scale normalized competency gap analysis across 10 career tracks, missing vs weak skills.
Skill ROI	Phase 11.1 Skill ROI Engine	Implemented & Verified: Marginal readiness velocity, acquisition effort estimation, quick-win vs high-ROI rankings.
Portfolio Projects	Phase 11.2 Capstone Recommendations	Implemented & Verified: 10 curated end-to-end production capstone projects targeting specific student skill gaps.
Academic Benchmarking	Phase 11.3 Degree Syllabus Alignment	Implemented & Verified: Curriculum alignment against accredited degree syllabi, identifying industry blind-spots.
Industry Demand	Phase 11.4 Labor Market Growth	Implemented & Verified: Labor demand signals, compound annual growth rate indicators, vacancy saturation metrics.
Career Trajectory	Phase 11.5 Multi-Hop Graph Pathways	Implemented & Verified: Graph traversal for transition pathways, intermediate stepping-stone roles, bridge skills.
Interview Simulator	Phase 11.6 AI Mock Interview Engine	Implemented & Verified: 30+ structured technical/behavioral scenario questions, STAR rubrics, defense scoring.
Evaluation Metrics	Phase 12.1 Recommendation Evaluation	Implemented & Verified: Precision@K, Recall@K, NDCG, novelty, diversity metrics, benchmark validation.
Job Action Center	Phase 12.2 Live Vacancy Intelligence	Implemented & Verified: Vacancy aggregation, ATS skill alignment, job readiness scores, job-specific missing skills.
Practical Assessment	Phase 12.3 Hands-On Task Engine	Implemented & Verified: 30 canonical diagnostic scenario tasks, automated in-memory rubric scoring without code exec.
Adaptive Curriculum	Phase 12.4 Adaptive Learning Engine	Implemented & Verified: 6-factor composite priority ranking, prerequisite DAG resolution, 5-stage learning progression.
Action Plan & Readiness	Phase 12.5 Readiness & Action Center	Implemented & Verified: What to do Today, Next, Blockers, Job-readiness, 5-day custom schedules (5h, 10h, 15h, 20h).
Third-Party SSO	OAuth Google / GitHub / LinkedIn	Future Scope: Direct OAuth 2.0 social login (currently authenticated via JWT email/password).
Live Scrapers	External Paid Labor Scraping APIs	Future Scope: Real-time external job board scraping (currently uses curated labor vacancy datasets with provenance).
Voice Mock Interviews	Real-Time WebRTC Audio/Video LLM	Future Scope: Speech-to-text live interview recording with LLM voice synthesis.

7. Final Verification & Test Regression Results

Verification Suite	Command Executed	Tests / Result	Duration / Status
Phase 12.5 Unit Tests	python -m unittest backend/tests/test_phase12_module5.py	45 / 45 PASS	20.18s 100% PASS
Phase 12.4 Unit Tests	python -m unittest backend/tests/test_phase12_module4.py	33 / 33 PASS	14.00s 100% PASS
Phase 12.3 Unit Tests	python -m unittest backend/tests/test_phase12_module3.py	36 / 36 PASS	11.23s 100% PASS
Master Full Regression	python -m unittest discover -s backend/tests	471 / 471 PASS	124.10s ZERO FAILURES
Frontend Production Build	npm --prefix frontend run build	Exit Code 0	10.13s VITE SUCCESS
Database Schema Audit	inspect(db.engine).get_table_names()	11 Tables Intact	0 Schema Mutations
Git Formatting Check	git diff --check	Clean Diff	0 Whitespace Errors

8. Final Project / Git Checkpoint Reference

Repository Path: /home/sumit/Desktop/projects/AI-Career-Navigator
Active Branch: main
Protected Base Commit: e88b7e97c2c9b013855e997d5f63e517c5b06269 (feat: add practical skill assessment engine)
Working Tree Status: Contains uncommitted Phase 12.4 and Phase 12.5 enhancements ready for review. Strictly preserved in accordance with user safety rules (no commit, push, reset, restore, checkout, clean, or stash executed).
Verification Conclusion: All 471 unit tests passing, frontend production bundle built cleanly, live daemons running on ports 5000 and 5173.